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Weathering the storm: inter-municipal cooperation and citizens' perceptions of flood preparedness

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Abstract

We study whether the provision of specific functions through inter-municipal cooperation (IMC), rather than by single municipalities, affects citizens' perceptions of local government management and service delivery. We use the floods that affected Emilia-Romagna in May 2023 as a natural experiment to examine whether citizens living in municipalities that cooperated through inter-municipal arrangements evaluated local flood management differently from those living in municipalities providing services independently. We combine survey data collected after the disaster with administrative information on municipal service cooperation. Our results show that, in flood-affected municipalities, the probability that citizens report extreme dissatisfaction with prevention measures increases by 24.6 percentage points in non-IMC municipalities compared to municipalities not affected by floods. This probability falls to 13.4 percentage points in municipalities participating in IMC arrangements, indicating that IMC attenuates the negative effect of the disaster on citizens' evaluations by 11.2 percentage points. A channel decomposition reveals this protective effect is entirely driven by joint territorial management — the cooperative function covering land-use planning and environmental services — while IMC in all other services shows no effect. Our findings suggest that cooperation in functions directly related to disaster prevention and territorial governance can shield local governments from the reputational costs of adverse shocks.

JEL Codes: H70, H72, Q54, R58

Keywords: Inter-municipal cooperation, Flood preparedness, Citizen perceptions, Natural disasters

1. Introduction

Natural disasters place local governments on the front line of crisis management. Municipalities are responsible for many of the functions that shape both disaster preparedness and emergency response, including land-use planning, environmental management, infrastructure maintenance, and civil protection. When a flood, wildfire, or other environmental shock occurs, citizens experience these institutions as the closest and most visible level of government, making local authorities particularly exposed to public scrutiny. Because many prevention and territorial management functions are organized at the municipal level, natural disasters provide a unique opportunity to evaluate whether different institutional arrangements affect citizens' perceptions of government performance. The literature has already investigated that beyond physical damage, severe environmental shocks impose a political cost: citizens revise their evaluations of local governments based on how well the crisis was anticipated and managed (Gasper & Reeves, 2011; Healy & Malhotra, 2009). Whether institutional arrangements — rather than individual competence or fiscal capacity — can buffer this political cost is largely unexplored.

Inter-municipal cooperation (IMC) has been studied primarily in peacetime settings, where it may generate scale economies, reduce costs, or professionalize service delivery (Bel & Warner, 2015; Ferraresi et al., 2018). Whether these organizational gains translate into resilience during crises, and whether they are visible to citizens, is an open question with direct implications for the design of local government institutions.

We answer it using the May 2023 Emilia-Romagna (ER) floods — one of the most severe hydrological events in recent Italian history — as a natural experiment. Combining a post-disaster citizen survey with administrative data on municipal service cooperation (SOSE, 2022), we estimate the moderating effect of pre-existing IMC arrangements on citizen satisfaction with both prevention efforts and emergency management.

We test three IMC definitions: general participation in any cooperative service, participation specifically in joint territorial management (land-use planning, green infrastructure, waterway maintenance), and a residual measure — IMC excluding territorial management. Our main findings are: (i) flood exposure raises extreme dissatisfaction with prevention by 24.6 pp in non-IMC municipalities compared to municipalities not exposed to flood but only 13.4 pp in IMC ones — a gap of 11.2 pp; (ii) this difference is entirely explained by joint territorial management — IMC in all other services shows no effect.

2. Literature Review

Natural disasters impose a dual burden on local governments: a physical one measured in infrastructure damage, and a political one through which citizens revise their evaluations of the institutions responsible for managing risk. The political economy literature establishes that voters systematically reward post-disaster relief spending while remaining largely unresponsive to preparedness investments, even though preparedness generates far larger social returns (Healy & Malhotra, 2009). The mechanism is partly observational — relief disbursements are visible, attributable, and immediate, while the benefits of prevention are invisible — citizens never see the disaster that was averted — and partly temporal: the official who invested in floodplain management years earlier receives none of the political credit when the levee holds. Gasper and Reeves (2011) refine this picture: disaster damage does hurt incumbents at the polls, but citizens respond most strongly to attributable institutional action (who declared the emergency, who deployed resources). When functional responsibility is clear, citizens can distinguish competent management from bad luck. European evidence is more muted. Albrecht (2017) finds that political trust and government satisfaction are largely stable across ten disaster cases, shifting meaningfully only when management is visibly perceived to have failed. For our study, this means that citizens' post-disaster evaluations depend on how well institutions are perceived to have performed, not just on the physical extent of the damage.

Effective disaster performance is inherently organizational. Boin and 't Hart (2003) document that prevention and preparedness are politically unrewarding tasks — successful mitigation generates no credit and no media attention — making leaders prone to underinvest in prevention, with crisis response left to improvisation. Waugh and Streib (2006) show that modern emergency management depends on pre-existing polycentric networks: prevention — land-use regulation, waterway maintenance, hazard mitigation — is a local, inter-organizational responsibility, and communities whose governments maintain collaborative relationships before a disaster strikes are better positioned to limit damage and sustain response capacity.

Inter-municipal cooperation (IMC) is a candidate institutional form for building exactly this kind of pre-existing collaborative capacity at the local level. Yet it has been studied almost exclusively as a peacetime efficiency instrument. The economics literature documents consistent cost savings from voluntary cooperation (Bel & Warner, 2015; Ferraresi et al., 2018), with causal estimates for Italian *Unioni di Comuni* confirming efficiency gains (Luca & Modrego, 2021). More recent contributions extend the picture: Gori et al. (2025) show that cost reductions do not automatically translate into better ser-

vices; Elston et al. (2025) find citizen-facing quality improvements in English housing allowances; Notsu et al. (2025) document strengthened tax enforcement capacity. Voluntary cooperation generally yields net gains (Bel & Warner, 2015), whereas coerced arrangements can impose costs (Tricaud, 2025), and multi-principal governance introduces accountability trade-offs (Voorn et al., 2019).

The two bodies of evidence share a gap. The disaster literature treats institutional structure as given; the IMC literature examines steady-state performance in normal times. Neither asks what happens when a pre-existing cooperative arrangement is stress-tested by an acute shock, nor whether it shapes citizens' evaluations of their local government in the aftermath. We sit at this intersection: exploiting the 2023 Emilia-Romagna floods as a quasi-natural experiment, we estimate whether inter-municipal cooperation moderates the political cost of disaster exposure as perceived by citizens.

3. Institutional Background

In May 2023, two consecutive extreme rainfall events struck the Italian region of Emilia-Romagna. Twenty-three rivers burst their banks, directly flooding approximately 540km^2 across the provinces of Ravenna, Forlì-Cesena, Bologna, and Rimini. The event caused 17 deaths, displaced roughly 36,000 people, and produced an estimated 8.9 billion euros in damages (see the document from *Camera dei Deputati VIII Commissione Ambiente, Territorio e Lavori pubblici*). The spatial distribution of flood exposure was primarily determined by pre-existing hydrological and geomorphological characteristics, providing the exogenous shock we exploit.

Italian municipalities may pool administrative services through Inter-Municipal Cooperation (IMC), formal cooperative arrangements in which member municipalities jointly manage selected public functions.¹ Participation is largely voluntary; specific services covered vary across unions and are documented in the annual SOSE survey. For municipalities below 5,000 inhabitants, the 2010 Spending Review mandated cooperation for a set of core functions, though enforcement has been uneven. The function relevant to our study is Joint Territorial Management (JTM, in Italian *gestione del territorio*): land-use planning, urban green spaces, environmental monitoring, and park management. These activities directly shape both pre-disaster prevention infrastructure (floodplain zoning,

¹Under the *Testo Unico degli Enti Locali* (TUEL, D.Lgs. 267/2000), the two main legal instruments are the *Convenzione* (Art. 30, a contractual agreement that does not create a new legal entity) and the *Unione di Comuni* (Art. 32, an autonomous public body with its own budget and organs). We treat both as IMC throughout.

green buffer zones) and baseline organizational capacity for environmental emergencies.²

4. Data

Our primary data are a post-disaster citizen survey conducted in ER municipalities after the 2023 floods, merged with two administrative sources: (i) SOSE (2022) for municipal IMC service agreements; (ii) ISTAT for municipal structural controls. A follow-up survey wave collected emergency management ratings several months after the event. The analysis dataset contains 2,062 respondents across 259 municipalities.

From the survey, we use two main outcomes: perceived prevention efficacy and emergency management rating. The first captures citizens' assessments of the municipality's pre-disaster prevention efforts, while the second captures their evaluation of the local government's crisis response. Both are measured at the individual level through survey questions. For the prevention outcome, we use the question: "*Did your municipality carry out sufficient prevention activities (e.g., the construction of containment basins)?*" with a four-point Likert scale (1 = Not at all, 4 = Very much). For the emergency management outcome, we use: "*Using a scale from 0 to 10, how would you rate your municipality's management of the May 2023 flood?*" with 0 = Worst possible and 10 = Best possible.

We also have a binary variable for flood exposure, equal to one if the municipality was flooded and zero otherwise, and a set of individual-level controls capturing demographic characteristics³.

From the SOSE dataset, we construct the treatment variables capturing inter-municipal cooperation. The SOSE survey collects detailed information on which public services are managed jointly by each municipality and its neighbors. We use this to construct three key variables: (i) a general IMC indicator equal to one if the municipality participates in any cooperative service; (ii) a specific indicator for joint territorial management (JTM); and (iii) a residual IMC variable capturing cooperation in any service except territorial management.

4.1. Descriptive Statistics

The sample includes 1,248 respondents in non-flooded municipalities and 814 in flooded ones (60.5% and 39.5%); 890 respondents (43.2%) are in IMC municipalities, of which

²"Joint Territorial Management" covers the SOSE service category *Territorio, ambiente, urbanistica*.

³age, gender, education, employment status, income bracket, municipal population, baseline political attitudes, trust in the local mayor, and self-reported municipal damage

441 live in a municipality with active Joint Territorial Management (JTM). Table 1 reports summary statistics.⁴ Flooded municipalities report lower prevention efficacy and emergency management ratings on average; among flooded municipalities, IMC municipalities score higher on both outcomes. IMC municipalities are smaller on average, while flooded municipalities are larger than non-flooded ones.

Table 1
Descriptive Statistics by Flood Exposure and IMC Status

	Not Flooded		Flooded	
	No-IMC	IMC	No-IMC	IMC
<i>Outcomes and treatment</i>				
Joint Territorial Management (JTM)	0.000 (0.000)	0.355 (0.479)	0.000 (0.000)	0.715 (0.452)
Prevention efficacy (1–4)	2.232 (0.949)	2.128 (0.911)	1.964 (0.830)	2.007 (0.797)
Emergency management (0–10)	5.766 (2.433)	6.128 (2.438)	5.032 (2.672)	5.855 (2.458)
<i>Individual-level controls</i>				
Age	48.396 (14.335)	46.112 (14.135)	48.734 (14.877)	47.648 (15.155)
Female	0.496 (0.500)	0.488 (0.500)	0.445 (0.498)	0.530 (0.500)
High school diploma	0.492 (0.500)	0.547 (0.498)	0.439 (0.497)	0.501 (0.501)
University degree	0.423 (0.494)	0.293 (0.455)	0.473 (0.500)	0.360 (0.481)
Unemployed	0.060 (0.237)	0.068 (0.252)	0.045 (0.207)	0.058 (0.233)
Employed	0.606 (0.489)	0.619 (0.486)	0.593 (0.492)	0.550 (0.498)
Income <20k (ref.)	0.152 (0.359)	0.192 (0.394)	0.186 (0.390)	0.242 (0.429)

(continued on next page)

⁴Balance tests comparing observable pre-treatment characteristics by IMC status and flood status are reported in Appendix Tables A1 and A2.

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	Not Flooded		Flooded	
	No-IMC	IMC	No-IMC	IMC
Income 20–40k	0.418 (0.494)	0.449 (0.498)	0.467 (0.499)	0.412 (0.493)
Income 40–60k	0.201 (0.401)	0.123 (0.329)	0.158 (0.366)	0.153 (0.360)
Income 60–80k	0.065 (0.247)	0.033 (0.179)	0.041 (0.198)	0.023 (0.150)
Income >80k	0.023 (0.149)	0.022 (0.147)	0.049 (0.217)	0.017 (0.131)
Trust in mayor	0.462 (0.499)	0.506 (0.500)	0.514 (0.500)	0.504 (0.501)
<i>Municipal-level controls</i>				
Population	1.07e+05 (75724.189)	18367.961 (17061.583)	2.83e+05 (1.42e+05)	27560.839 (28478.151)

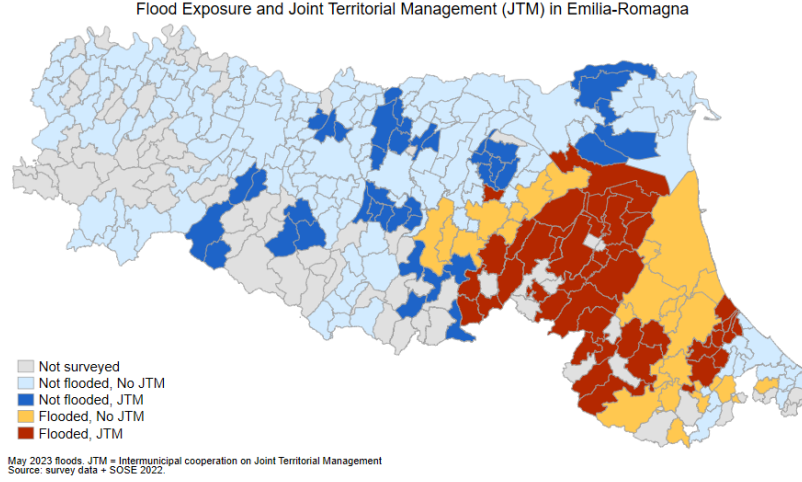
Note: Standard deviations in parentheses.

A key preliminary check concerns differential damage. Among flooded respondents, the mean perceived municipal damage score (lower = more damage) is 2.10 for IMC municipalities versus 2.36 for non-IMC. IMC municipalities were on average more severely hit. Therefore, any positive effect of IMC on satisfaction therefore reflects institutional quality, not lighter physical exposure. A complementary check supports the validity of the prevention measure: among flooded respondents, those who rate prevention more positively also report lower municipal damage, suggesting that prevention efforts made a real physical difference and that the survey question captures something meaningful.

Figure 1 shows the geographic distribution of municipalities in the sample by flood exposure and municipalities with Joint Territorial Management. Flooded municipalities are concentrated in the provinces of Ravenna, Forlì-Cesena, and Bologna. IMC municipalities are distributed across the region (dark blue and red), concentrating in rural areas of the Apennine mountains but also including some municipalities in the plains and near the coast.

Figure 1

Emilia-Romagna municipalities by flood exposure and Joint Territorial Management status.



5. Empirical Strategy

Our specification exploits flood exposure as the treatment and examines whether its effect on citizens' evaluations differs between municipalities with and without pre-existing inter-municipal cooperation arrangements. In this sense, IMC is the institutional condition along which we estimate heterogeneous effects. The identifying comparison is between flooded and non-flooded municipalities, allowing the effect of the shock to vary by whether municipalities managed relevant functions individually or through IMC.

$$Y_{ic}^* = \beta_0 + \beta_1 Flood_c + \beta_2 IMC_c + \beta_3 (Flood_c \times IMC_c) + \gamma_1' X_{ic} + \gamma_2' W_{ic} + \gamma_3 Pop_c + \varepsilon_{ic}$$

Let Y_{ic}^* be the latent satisfaction of citizen i in municipality c . $Flood_c$ is a binary variable equal to one if municipality c was flooded and zero otherwise; IMC_c is a binary variable equal to one if municipality c participates in inter-municipal cooperation and zero otherwise; X_{ic} is a vector of individual-level controls (age, gender, education, income, job); W_{ic} is a vector of two variables from the survey (trust in the mayor and self-reported damage); Pop_c is the population of municipality c . The error term ε_{ic} captures unobserved factors affecting satisfaction.

The observed outcome Y_{ic} is an ordinal variable derived from Y_{ic}^* , corresponding to the survey response categories and referring to the prevention efficacy and emergency management (see Section 4). Given the ordinal scale of both outcomes, we estimate ordered probit models with standard errors clustered at the province level to account for within-province correlation.

The parameter of interest is β_3 : it tells us whether the effect of the flood on citizens' satisfaction is different in IMC and non-IMC municipalities. Identification rests on two assumptions. First, flood exposure is plausibly exogenous: as discussed in Section 3, the spatial distribution of flooding was determined by pre-existing hydrological and geomorphological characteristics unrelated to municipal institutional arrangements. Second, IMC status is predetermined: cooperation agreements are recorded in the 2022 SOSE survey, prior to the disaster, ruling out reverse causality in the interaction term. Together, these conditions support a causal interpretation of $\hat{\beta}_3$ as the heterogeneous treatment effect of flood exposure across different institutional settings.

We run three specifications varying IMC_c : general cooperation (IMC_c), joint territorial management (IMC_c^{terr}), and residual IMC (IMC_c^{res} , i.e. IMC in any service except territorial management). The residual specification serves as a falsification test: if the protective effect operates through territorial management capacity, cooperation in all other services should produce no significant interaction.

6. Results

6.1. Prevention Efficacy

Figure 2 shows how flood exposure shifts the probability of each satisfaction level, separately for IMC (red) and non-IMC municipalities (blue). Both groups suffer from the flood — being in a flooded area always worsens prevention perceptions — but by very different amounts.

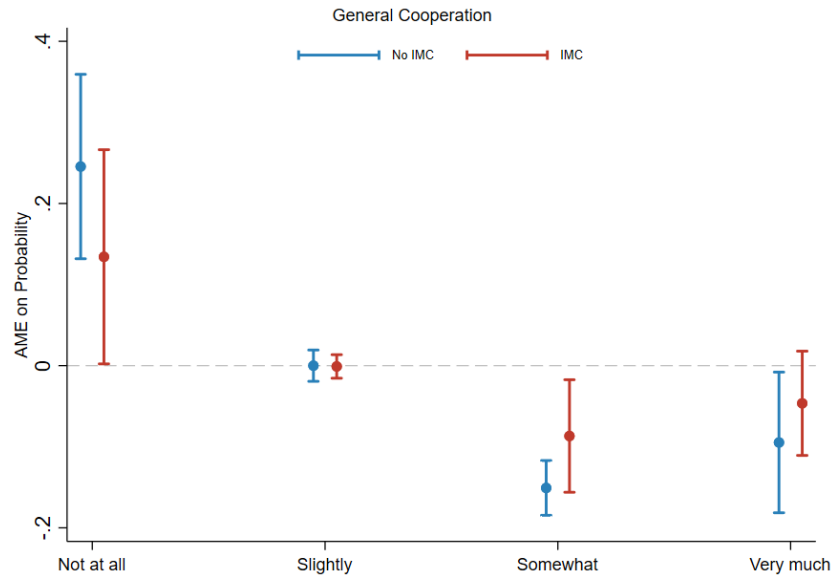
For non-IMC municipalities, the flood substantially increases dissatisfaction with prevention measures: the probability of reporting the lowest rating ("Not at all") rises by 24.6 percentage points ($p < 0.001$). By contrast, in IMC municipalities the same shock produces a much smaller increase of 13.4 percentage points ($p = 0.046$). This implies that the flood-induced increase in extreme dissatisfaction is 11.2 percentage points lower in IMC municipalities than in non-IMC municipalities, a difference that is statistically significant ($\hat{\beta}_3 = 0.399$, $p < 0.001$).⁵ Importantly, IMC participation has no significant effect in non-flooded municipalities, suggesting that the difference emerges only after the disaster rather than reflecting systematically different baseline evaluations. Moreover, among flooded municipalities, respondents in IMC municipalities report slightly more severe damage on average than those in non-IMC municipalities (mean damage index 2.10 versus 2.36, where lower values indicate greater damage). The attenuation effect associ-

⁵Full ordered probit coefficient estimates for the prevention outcome, including threshold parameters, are reported in Appendix Table A5.

ated with IMC therefore does not appear to be driven by lighter exposure to the flood and, if anything, is observed despite greater reported damage.

Figure 2

Average marginal effect of flood exposure on perceived prevention efficacy, by IMC status. Each point is the change in probability of each satisfaction rating caused by being in a flooded municipality, separately for municipalities without (blue) and with (red) inter-municipal cooperation. Points above zero: the flood raises that rating's probability. Ordered probit. Standard errors clustered at province level. $N = 1,564$.



6.2. Channel Decomposition

Figure 3 disaggregates the protection effect by running the same ordered probit model twice, each time replacing the IMC indicator: joint territorial management (left) and residual IMC, all services except territorial management (right). The format is identical to Figure 2: each panel shows the AME of flood exposure on each outcome level, separately for municipalities with (red) and without (blue) the relevant cooperative function.⁶

The panels jointly identify the mechanism. Municipalities with JTM show a large and significant flood interaction ($\hat{\beta}_3 = 0.507$, $p < 0.001$): the red and blue lines diverge

⁶We additionally estimated a specification using civil protection (CP) IMC as the treatment variable and found a significant interaction. However, JTM is perfectly nested inside CP in the data: every municipality with JTM also cooperates on civil protection, with no municipality holding JTM without CP. The bivariate CP coefficient therefore conflates municipalities with and without JTM. A categorical specification separating CP-only from JTM+CP municipalities confirms that the protective effect belongs entirely to the JTM+CP group; CP-only municipalities are not significantly shielded. Civil protection membership alone provides no reputational buffer.

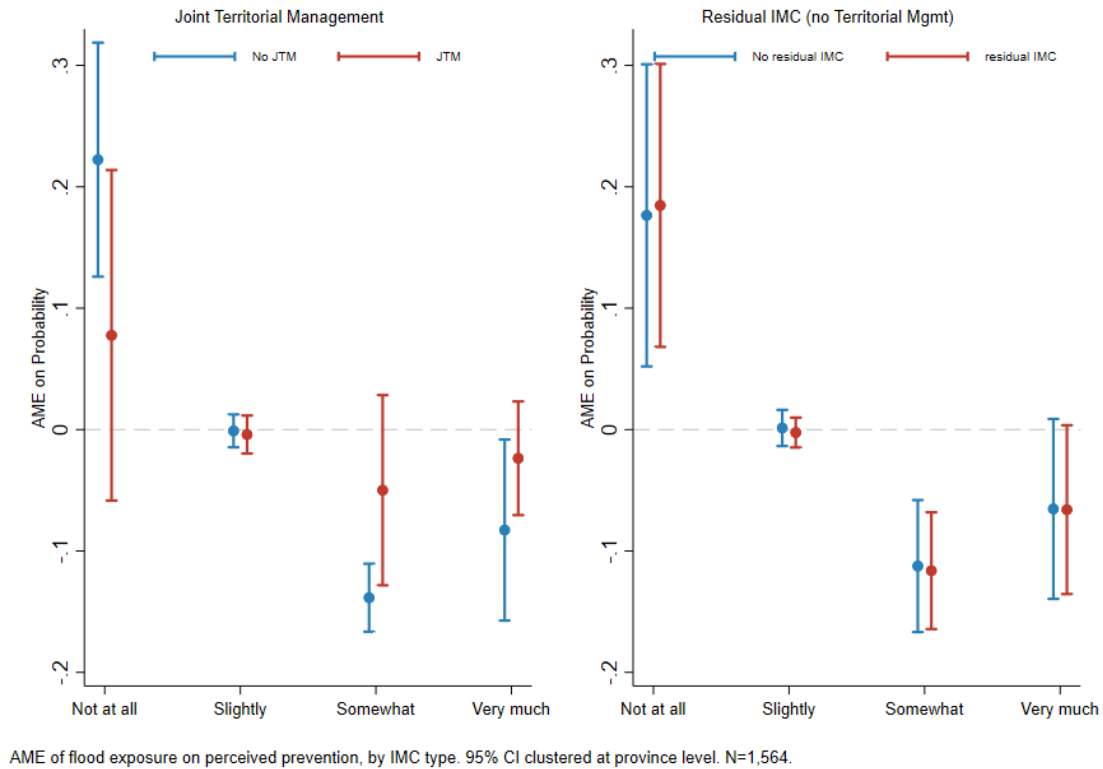
sharply, with JTM municipalities suffering a markedly smaller shift toward the worst outcome. One ancillary finding is worth noting: the main effect of JTM among non-flooded municipalities is negative and significant ($\hat{\beta}_2 \approx -0.21$, $p < 0.05$), a peacetime penalty. The most natural reading is reverse selection: municipalities facing more demanding environmental conditions — more complex waterway systems, greater flood exposure — are precisely those that choose to cooperate on territorial management. Citizens in these municipalities hold higher baseline expectations of prevention effort, and rate it somewhat more critically in normal times. When a major flood does strike, those same institutional arrangements translate into a measurable buffer. By contrast, municipalities not in IMC, or cooperating in any service *except* territorial management show no protective effect: the red and blue lines are virtually identical ($\hat{\beta}_3 = -0.023$, $p = 0.882$), a clean falsification. The general IMC (see Figure 2) inherits the protective effect because the general IMC category includes JTM municipalities; once these are separated out, all the explanatory power resides in territorial management capacity alone.

Restricting each specification to its treatment group and pure No-IMC municipalities, removing contamination of the reference group by other IMC types, strengthens both results: the JTM interaction rises to $\hat{\beta}_3 = 0.607$ ($p < 0.001$) and the residual IMC interaction remains indistinguishable from zero ($\hat{\beta}_3 = 0.200$, $p = 0.296$). Appendix Figure A2 shows the restricted-sample decomposition.⁷

⁷To stress these results, we also applied an inverse probability weighting (IPW) on municipal-level observables to account for selection into IMC. The results are robust, see Appendix Figure A3.

Figure 3

Channel decomposition: average marginal effect of flood exposure on perceived prevention efficacy, by IMC specification. Left: joint territorial management (JTM). Right: residual IMC (all services except territorial management). Blue: municipalities without the relevant cooperative function. Red: municipalities with the relevant cooperative function. Each point is the change in probability of each satisfaction rating caused by being in a flooded municipality. Ordered probit. Standard errors clustered at province level.



6.3. Emergency Management

Prevention and emergency management are conceptually related dimensions of local disaster governance. Institutions that cooperate in territorial management before a shock may be perceived as better prepared to reduce local vulnerability and to coordinate the subsequent emergency response. We therefore treat the two outcomes as complementary rather than interchangeable: prevention efficacy captures citizens' assessments of pre-disaster preparedness, whereas emergency management reflects their evaluation of the municipality's response after the flood. Examining both outcomes allows us to assess whether the role of IMC extends across different stages of the disaster cycle.

Figure 4 shows the average marginal effect of flood exposure on the emergency management rating (0–10 scale, $N = 1,189$), by IMC status, for all three specifications. To simplify interpretation, we recode the original 0–10 emergency management rating into

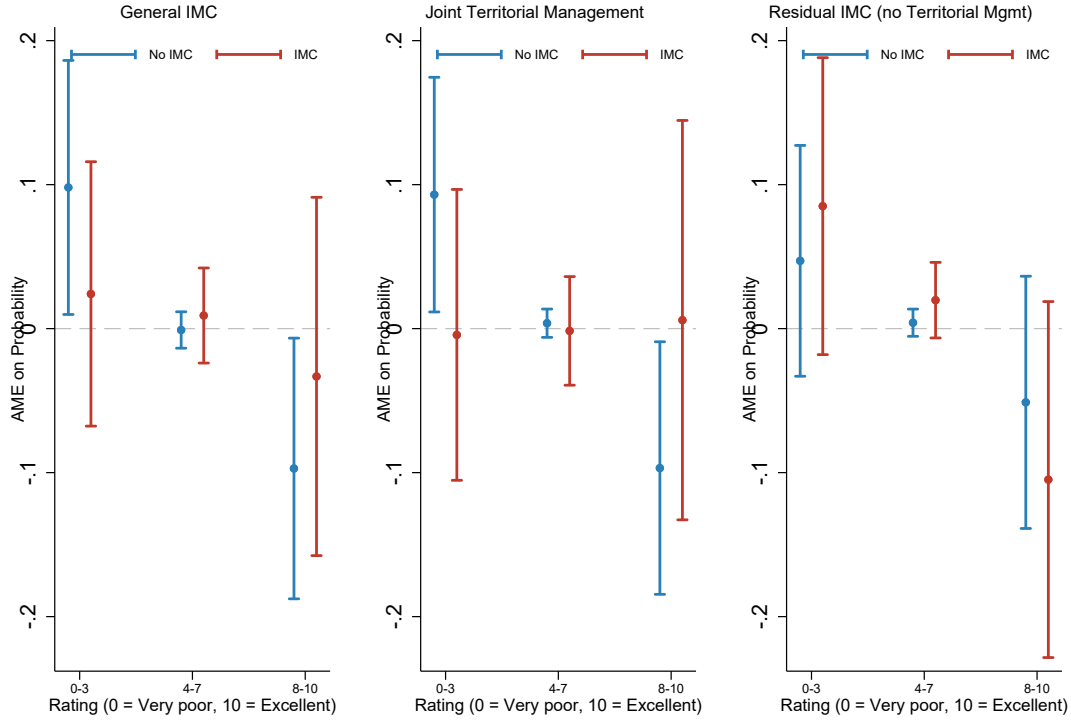
three categories: low ratings (0–3), medium ratings (4–7), and high ratings (8–10).

Raw means support the hypothesis directionally: flooded non-IMC municipalities average 5.03, flooded IMC municipalities 5.85 (non-flooded: 5.77 vs. 6.13). The results move in the expected direction. In municipalities participating in general IMC or Joint Territorial Management (JTM), flood exposure is associated with a smaller increase in the probability of low ratings and a smaller decline in the probability of high ratings. However, these differences are estimated imprecisely and the interaction coefficients are not statistically significant.⁸ The residual IMC specification provides a useful comparison. Municipalities cooperating in services other than territorial management display no evidence of a protective effect, with interaction coefficients close to zero and statistically insignificant. This clean null is consistent with the prevention analysis: the positive role of IMC appears to be concentrated in territorial management rather than in cooperation per se.

⁸Full ordered probit coefficient estimates for the emergency management outcome are reported in Appendix Table A6.

Figure 4

Average marginal effect of flood exposure on emergency management rating (0–10 scale), by IMC status. Left to right: general IMC, joint territorial management, residual IMC. Each point is the AME of flood exposure on the probability of each rating category, for municipalities with (red) and without (blue) the relevant cooperative function. Ordered probit. $N = 1,189$.



AME of flood exposure on emergency management rating, by IMC type. 95% CI clustered at province level. $N=1,189$.

7. Robustness and Placebo Tests

Since we cannot exploit a pre-disaster baseline, we rely on two indirect pieces of evidence supporting our identification strategy. First, among non-flooded municipalities, the IMC coefficient is close to zero and statistically insignificant, suggesting that citizens in IMC and non-IMC municipalities report similar evaluations in the absence of the shock. This is a placebo check: without the flood shock, IMC predicts no difference in prevention evaluations, which supports the idea that the interaction effect captures disaster-specific institutional performance rather than pre-existing differences. Second, as discussed in Section 3, flood exposure was driven by hydrological and geomorphological characteristics of the territory rather than by municipalities' institutional arrangements. This makes flood exposure plausibly exogenous to IMC status and reduces the concern that the estimated interaction effect is driven by selection into cooperation. We test the sensitivity of our results to alternative specifications. We re-estimate the prevention efficacy model using: (i) a probit on the binary worst-outcome indicator (Not at all = 1, else 0); (ii)

OLS treating the four-point scale as continuous; and (iii) OLS with province fixed effects, absorbing all province-level confounders and identifying solely off within-province variation. The main findings survive across all three alternative specifications (Table 2).

The probit on the binary worst-outcome indicator confirms that IMC attenuates the flood penalty at the extreme end of the satisfaction distribution, without imposing the threshold assumptions of the ordered probit. Flood exposure raises P(Not at all) by 18.9 pp in non-IMC municipalities but only 5.3 pp in IMC ones — roughly a halving of the effect that mirrors the main results. For territorial IMC the pattern is similar (14.9 pp vs. 5.4 pp); residual IMC shows no meaningful attenuation (12.7 pp vs. 4.2 pp), maintaining the channel-specificity found throughout.

Treating the four-point scale as continuous, OLS yields significant positive interactions for general IMC ($\hat{\beta}_3 = 0.283$, $p < 0.01$) and territorial management ($\hat{\beta}_3 = 0.369$, $p < 0.01$), while the residual coefficient remains at -0.028 and is not significant. The same selectivity — territorial cooperation buffers, other cooperation does not — replicates exactly.

Adding province fixed effects is a more demanding test: it absorbs all province-level confounders, including differences in baseline flood exposure and institutional history, identifying solely off within-province variation. The general IMC interaction attenuates to 0.161 ($p = 0.11$), but the territorial management interaction remains highly significant ($\hat{\beta}_3 = 0.295$, $p < 0.01$). The fact that the territorial estimate holds within provinces strengthens the causal interpretation: the protective effect is not driven by province-level sorting between cooperating and non-cooperating municipalities.

Table 2

Robustness checks: prevention efficacy. Province-clustered standard errors in parentheses.
N = 1,564.

Specification	General IMC		Joint Territorial Mgmt		Residual IMC	
	No-IMC	IMC	No-JTM	JTM	No-Res.	Residual
Probit, P(Not at all)	0.189*** (0.063)	0.053 (0.037)	0.149*** (0.043)	0.054 (0.045)	0.127** (0.051)	0.042 (0.056)
OLS, score 1–4	−0.612*** (0.180)	0.283*** (0.076)	−0.551*** (0.155)	0.369*** (0.104)	−0.433** (0.179)	−0.028 (0.106)
OLS + Province FE	−0.244** (0.078)	0.161 (0.101)	−0.238** (0.073)	0.295*** (0.082)	−0.066 (0.058)	−0.152 (0.127)

Continued on next page

Specification	General IMC		Joint Territorial Mgmt		Residual IMC	
	No-IMC	IMC	No-JTM	JTM	No-Res.	Residual

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Province-clustered SEs in parentheses.

Probit: AME of flood on $P(\text{Not at all} = 1)$ by IMC status. OLS/FE: flood coefficient (No-IMC col.) and IMC×Flood interaction $\hat{\beta}_3$ (IMC col.). OLS+FE: province FE absorbed via areg.

Placebo outcomes. If the effect operates through local territorial management capacity, the IMC × Flood interaction should be absent for outcomes that do not depend on local institutional performance. We test three placebo outcomes using OLS with the general IMC specification: (i) satisfaction with central government financial aid to municipalities after the flood; (ii) regional government performance rating; (iii) general trust in politicians. All three interaction coefficients are small and statistically indistinguishable from zero: -0.168 for central aid, 0.085 for regional government, and -0.142 for political trust (all $p > 0.15$). The effect is specific to local prevention capacity, precisely the domain in which joint territorial management operates.

Table 3

Placebo tests: IMC × Flood interaction on outcomes unrelated to local prevention capacity. OLS with province-clustered standard errors. General IMC specification.

	Central gov. aid (Q068, scale 1–4)	Regional gov. rating (QF007, scale 0–10)	Trust in politicians (Q071)
Flood ($\hat{\beta}_{\text{Flood}}$)	0.052 (0.127)	-0.083 (0.332)	0.598^{**} (0.203)
IMC × Flood ($\hat{\beta}_3$)	-0.168 (0.131)	0.085 (0.365)	-0.142 (0.104)
N	1,314	1,189	2,062

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Province-clustered SEs in parentheses.

$\hat{\beta}_3 \approx 0$ throughout: the IMC buffering effect is specific to local prevention capacity.

8. Conclusion

This paper examines whether pre-existing inter-municipal cooperation shields local governments from the reputational costs of natural disasters. We exploit the May 2023 Emilia-Romagna floods as a natural experiment, combining a post-disaster citizen survey with administrative data on municipal service cooperation, and estimate ordered probit

models that interact flood exposure with three distinct inter-municipal cooperation (IMC) indicators. The design allows us to recover both the average protective effect and the specific institutional channel through which it operates.

The central finding is that flood exposure raises the probability of extreme dissatisfaction with prevention by 24.6 percentage points in non-IMC municipalities but only 13.4 points in IMC ones — a gap of 11.2 pp that survives individual-level controls, province fixed effects, and alternative functional forms. The mechanism is precise: the entire buffer resides in joint territorial management, the cooperative function covering land-use planning, environmental monitoring, and waterway maintenance. IMC in all other services yields an interaction indistinguishable from zero. The residual IMC specification provides a clean built-in falsification: if the protective effect operated through generic organizational capacity, any cooperation should help; it does not. On emergency management, the results for general IMC and JTM are directionally consistent but imprecise; the residual IMC estimate is indistinguishable from zero, reinforcing the interpretation that territorial capacity is the active ingredient.

Several limitations qualify these conclusions. The sample covers a single hydrological event in one Italian region, which limits external validity. We also cannot exclude that territorial management IMC proxies for broader governance quality correlated with flood preparedness. Finally, citizen satisfaction is a reputational outcome, not a direct measure of service quality: better-organized municipalities may communicate their capacity more effectively without necessarily performing better on the ground.

Despite these limitations, the findings carry a clear policy message. The large literature on inter-municipal cooperation has focused almost exclusively on peacetime efficiency gains — cost savings, administrative professionalization, scale economies. Our results extend this to crisis settings and show that the specific cooperative function matters more than participation in cooperation per se. From a policy design perspective, this argues for treating territorial management agreements as a component of climate-resilience planning, not merely as an administrative efficiency tool. As the frequency and severity of hydrological disasters increase under climate change, identifying which institutional arrangements buffer the political costs of natural disasters — and preserve citizens' trust in local government when it matters most — becomes an increasingly urgent question for both economists and policymakers.

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Appendix

Table A1

Balance: IMC vs. Non-IMC Municipalities

	No-IMC	IMC	Difference	<i>p</i> -value
Age	48.531	46.711	1.819**	0.005
Female	0.476	0.504	−0.028	0.202
Education (1–4)	3.349	3.155	0.194***	0.000
Employment (1–6)	2.706	2.690	0.016	0.781
Income (1–5)	2.724	2.721	0.003	0.966
Trust in mayor	0.483	0.506	−0.023	0.308
Population	176,972	21,952	155,020***	0.000

Note: Means of pre-treatment individual and municipal characteristics by IMC status. Education: ordinal 1–4 (higher = more educated). Employment: ordinal 1–6. Income: ordinal 1–5 (higher = higher income bracket). Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Two-sided *t*-test of equal means. $N = 2,062$.

Table A2

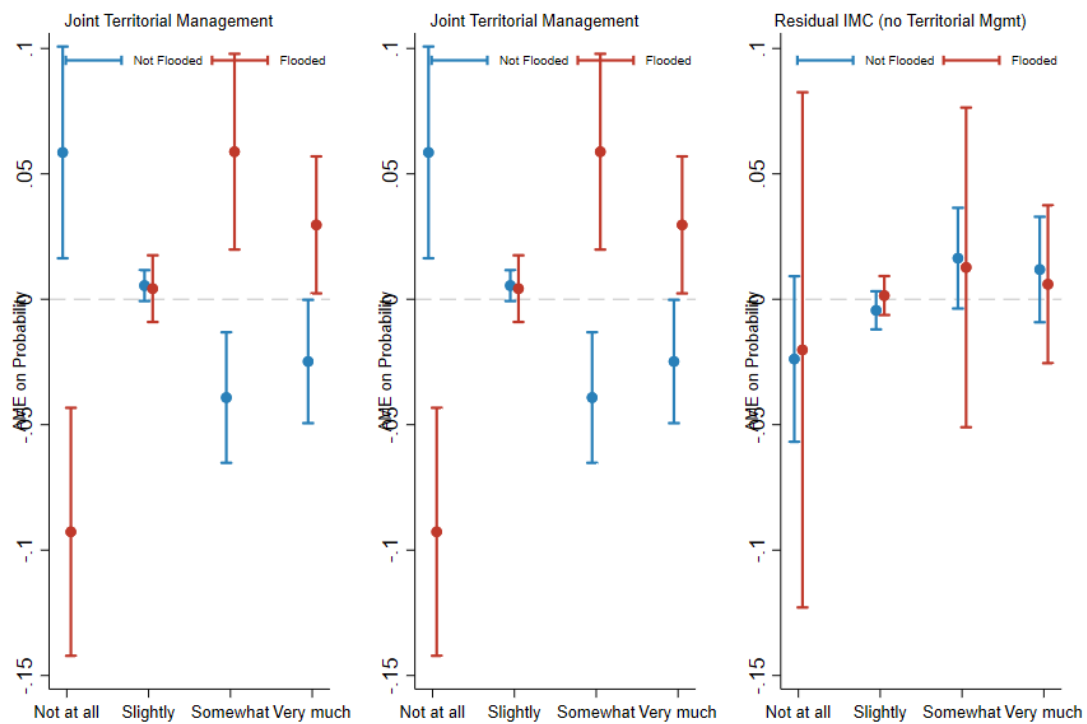
Balance: Flooded vs. Non-Flooded Municipalities

	Not flooded	Flooded	Difference	<i>p</i> -value
Age	47.402	48.271	−0.869	0.186
Female	0.493	0.482	0.011	0.619
Education (1–4)	3.241	3.302	−0.061*	0.046
Employment (1–6)	2.655	2.765	−0.110	0.055
Income (1–5)	2.800	2.606	0.194**	0.007
Trust in mayor	0.482	0.510	−0.028	0.210
Population	68,344	174,023	−105,679***	0.000

Note: Means of pre-treatment individual and municipal characteristics by flood status. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Two-sided *t*-test of equal means. $N = 2,062$.

Figure A1

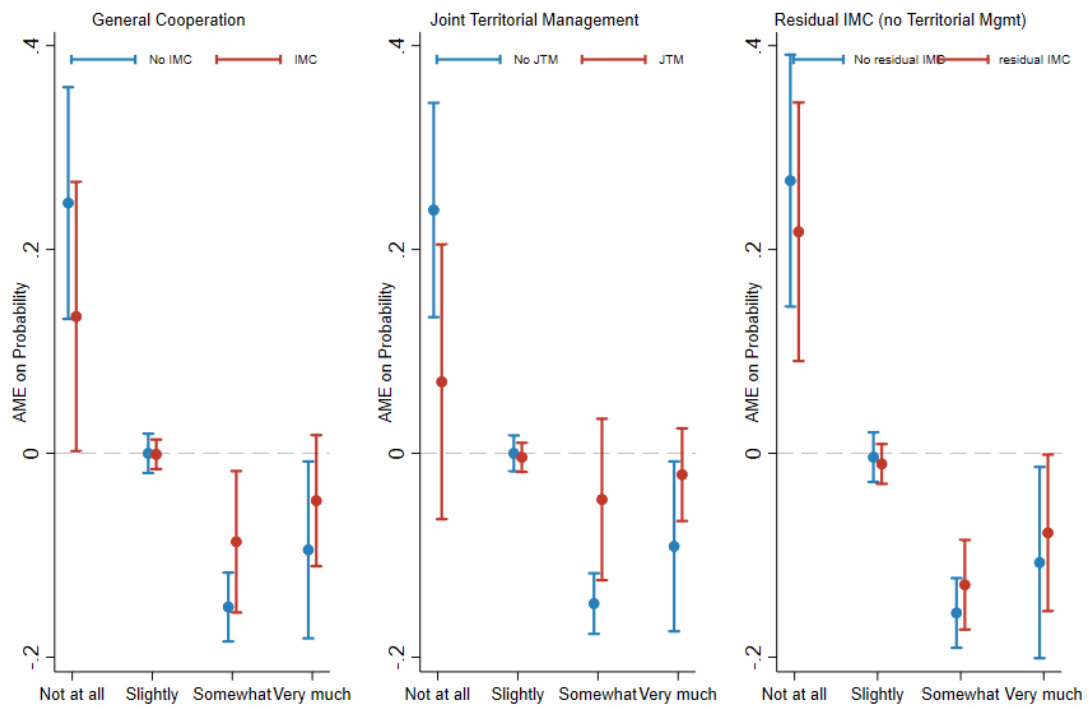
Margins of IMC type over perceived prevention efficacy, by flood status. Left: joint territorial management. Right: residual IMC (all services except territorial management). Each point is the change in probability of each satisfaction rating for municipalities with that cooperative function, separately for non-flooded (blue) and flooded (red). Ordered probit. Standard errors clustered at province level. N = 1,564.



AME of IMC type on perceived prevention, by flood status. 95% CI clustered at province level. N=1,564.

Figure A2

Channel decomposition (restricted samples): average marginal effect of flood exposure on perceived prevention efficacy. Left: general IMC, full sample ($N = 1,564$). Centre: JTM vs. pure No-IMC only ($N = 1,237$; CP-only and other IMC excluded). Right: residual IMC vs. pure No-IMC only ($N = 1,222$; JTM municipalities excluded). Blue: municipalities without the relevant cooperative function. Red: municipalities with the relevant cooperative function. Ordered probit. Standard errors clustered at province level.

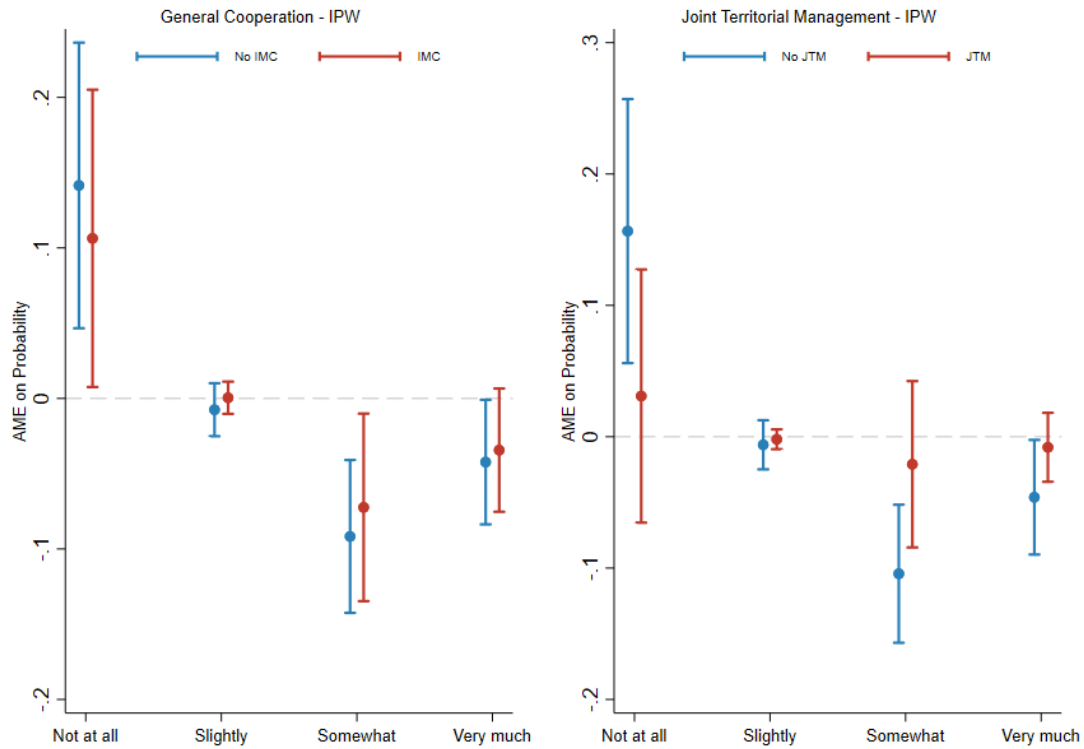


Left: General IMC, full sample. Centre: JTM vs. pure No-IMC. Right: Residual vs. pure No-IMC. AME of flood exposure on perceived prevention. 95% CI clustered at province level.

Inverse Probability Weighting. A potential concern is that municipalities selecting into inter-municipal cooperation may differ systematically from non-cooperating ones on observables correlated with flood preparedness and citizen evaluations. To assess robustness to this source of selection we implement ATT-stabilized inverse probability weighting (IPW). Propensity scores are estimated via logit at the municipality level using: log population, altitude, municipal area, urbanization type (two dummies), financial pressure index, tax pressure index, unemployment rate, and the share of population aged 65 and over. For general IMC the model fits well (pseudo- $R^2 = 0.13$), confirming meaningful variation in selection on observables. For JTM, we restrict estimation to the treatment group plus pure No-IMC municipalities ($n = 168$), excluding CP-only and other IMC types that would contaminate the counterfactual; within this restricted sample the propensity score achieves pseudo- $R^2 \approx 0.09$. Control units are reweighted using ATT stabilized Hájek-type weights ($w_i = \hat{p}_i / (1 - \hat{p}_i)$, normalized to sum to the number of controls) (Cunningham, 2021). Figure A3 shows the AME of flood exposure under IPW weighting. The pattern mirrors the main results: IMC municipalities suffer a substantially smaller flood-induced shift toward extreme dissatisfaction, and the JTM channel remains the active driver.

Figure A3

IPW robustness: average marginal effect of flood exposure on perceived prevention efficacy, by IMC status. Left: general IMC, IPW-weighted ordered probit ($N = 1,564$). Right: joint territorial management, restricted to pure No-IMC municipalities as control group, IPW-weighted ($N \approx 1,237$). Blue: municipalities without the relevant cooperative function. Red: municipalities with the relevant cooperative function. ATT stabilized Hájek weights. Standard errors clustered at province level.

**Table A3**

Average marginal effects of flood exposure on perceived prevention efficacy, by IMC status. Ordered probit with province-clustered standard errors. $N = 1,564$.

Outcome	General IMC		Joint Territorial Mgmt		Residual IMC	
	No-IMC	IMC	No-JTM	JTM	No-Res.	Residual
Not at all	0.246*** (0.058)	0.134** (0.067)	0.222*** (0.049)	0.078 (0.069)	0.176*** (0.063)	0.185*** (0.059)
Slightly	0.000 (0.010)	-0.001 (0.007)	-0.001 (0.007)	-0.004 (0.008)	0.001 (0.008)	-0.002 (0.006)
Somewhat	-0.151***	-0.087**	-0.139***	-0.050	-0.112***	-0.116***

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Outcome	General IMC		Joint Territorial Mgmt		Residual IMC	
	No-IMC	IMC	No-JTM	JTM	No-Res.	Residual
	(0.017)	(0.035)	(0.014)	(0.040)	(0.028)	(0.025)
Very much	−0.095**	−0.046	−0.083**	−0.024	−0.065*	−0.066*
	(0.044)	(0.033)	(0.038)	(0.024)	(0.038)	(0.036)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Province-clustered standard errors in parentheses.

AME of flood exposure (vs. not flooded) on probability of each prevention satisfaction category.

Scale: 1 = Not at all satisfied, 4 = Very much satisfied (higher = more satisfied).

Table A4

Average marginal effects of flood exposure on emergency management rating (0–10), by IMC status. Ordered probit with province-clustered standard errors. $N = 1,189$. Selected rating values shown.

Rating	General IMC		Joint Territorial Mgmt		Residual IMC	
	No-IMC	IMC	No-JTM	JTM	No-Res.	Residual
0 (worst)	0.045	0.011	0.040**	−0.003	0.020	0.035
	(0.039)	(0.020)	(0.016)	(0.020)	(0.022)	(0.037)
3	0.016***	0.006***	0.015***	−0.001	0.008***	0.017***
	(0.003)	(0.001)	(0.005)	(0.009)	(0.001)	(0.002)
5 (middle)	0.016	0.009**	0.017***	−0.002	0.009	0.024*
	(0.017)	(0.004)	(0.005)	(0.014)	(0.008)	(0.014)
8	−0.041***	−0.015***	−0.039***	0.004	−0.020***	−0.044***
	(0.004)	(0.002)	(0.015)	(0.025)	(0.001)	(0.005)
10 (best)	−0.028	−0.014	−0.029***	0.003	−0.015	−0.040
	(0.020)	(0.009)	(0.009)	(0.024)	(0.011)	(0.027)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Province-clustered standard errors in parentheses.

AME of flood exposure (vs. not flooded) on probability of each 0–10 rating value. Scale: 0 = Very poor, 10 = Excellent.

Table A5

Ordered probit estimates: perceived prevention efficacy. Province-clustered standard errors in parentheses. N = 1,564.

	(1) General IMC	(2) Territorial Mgmt	(3) Residual IMC
<i>Treatment variables</i>			
IMC	0.007 (0.109)		
JTM		−0.205*** (0.078)	
Residual IMC			0.087 (0.064)
Flood	−0.845*** (0.218)	−0.758*** (0.184)	−0.596*** (0.223)
IMC × Flood	0.399*** (0.101)		
JTM × Flood		0.507*** (0.137)	
Residual IMC × Flood			−0.023 (0.155)
<i>Individual controls</i>			
Age	−0.003 (0.002)	−0.003* (0.002)	−0.002 (0.002)
Female	−0.116 (0.074)	−0.120 (0.077)	−0.110 (0.076)
Educ: middle school	−0.331 (0.333)	−0.302 (0.330)	−0.332 (0.319)
Educ: high school	−0.366 (0.311)	−0.340 (0.314)	−0.364 (0.300)
Educ: university	−0.406 (0.375)	−0.378 (0.377)	−0.392 (0.363)
Empl: employed	0.277* (0.160)	0.277* (0.154)	0.278* (0.162)
Empl: self-employed	0.253 (0.220)	0.254 (0.212)	0.257 (0.221)

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	(1) General IMC	(2) Territorial Mgmt	(3) Residual IMC
Empl: student	0.267 (0.166)	0.267 (0.173)	0.290 (0.177)
Empl: retired	0.358* (0.190)	0.363** (0.185)	0.366* (0.188)
Empl: other	0.064 (0.192)	0.063 (0.192)	0.087 (0.197)
Income: 20–40k	0.058 (0.081)	0.058 (0.081)	0.049 (0.085)
Income: 40–60k	0.123 (0.143)	0.134 (0.144)	0.135 (0.145)
Income: 60–80k	0.457*** (0.150)	0.457*** (0.151)	0.463*** (0.153)
Income: >80k	0.251 (0.214)	0.267 (0.210)	0.240 (0.218)
Income: N/A	0.099 (0.154)	0.107 (0.152)	0.088 (0.158)
Trust in mayor	0.841*** (0.098)	0.850*** (0.098)	0.839*** (0.100)
<i>Municipal controls</i>			
Mun. damage: considerable	–0.004 (0.229)	–0.009 (0.233)	–0.015 (0.242)
Mun. damage: minor	–0.382 (0.372)	–0.388 (0.371)	–0.389 (0.380)
Mun. damage: none	–0.605 (0.402)	–0.609 (0.393)	–0.620 (0.412)
Home damage	–0.064 (0.082)	–0.064 (0.082)	–0.063 (0.084)
Population	0.000*** (0.000)	0.000*** (0.000)	0.000** (0.000)
<i>Threshold parameters</i>			
κ_1	–1.063*** (0.329)	–1.108*** (0.344)	–1.075*** (0.333)

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	(1) General IMC	(2) Territorial Mgmt	(3) Residual IMC
κ_2	0.160 (0.359)	0.115 (0.372)	0.143 (0.361)
κ_3	1.342*** (0.410)	1.300*** (0.428)	1.322*** (0.409)
Observations	1,564	1,564	1,564

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Province-clustered SE in parentheses.

Reference categories: primary education, unemployed, income <20k, heavy municipal damage.

Table A6

Ordered probit estimates: emergency management rating (3 categories: 0 = Low 0–3, 2 = Medium 4–7, 3 = High 8–10). Province-clustered standard errors in parentheses. N = 1,189.

	(1) General IMC	(2) Territorial Mgmt	(3) Residual IMC
<i>Treatment variables</i>			
IMC	0.056 (0.120)		
JTM		−0.024 (0.101)	
Residual IMC			0.029 (0.127)
Flood	−0.381** (0.181)	−0.371** (0.172)	−0.192 (0.167)
IMC × Flood	0.269 (0.201)		
JTM × Flood		0.391 (0.279)	
Residual IMC × Flood			−0.179 (0.166)
<i>Individual controls</i>			
Age	0.004 (0.003)	0.004 (0.003)	0.004 (0.003)
Female	−0.248***	−0.249***	−0.244***

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	(1) General IMC	(2) Territorial Mgmt	(3) Residual IMC
	(0.069)	(0.071)	(0.072)
Educ: middle school	0.112	0.127	0.115
	(0.423)	(0.410)	(0.417)
Educ: high school	0.336	0.346	0.330
	(0.339)	(0.336)	(0.336)
Educ: university	0.483	0.494	0.492
	(0.438)	(0.435)	(0.437)
Empl: employed	0.346	0.363	0.348
	(0.228)	(0.231)	(0.232)
Empl: self-employed	0.321	0.340	0.323
	(0.249)	(0.259)	(0.255)
Empl: student	0.312	0.346	0.344
	(0.341)	(0.343)	(0.352)
Empl: retired	0.591**	0.609**	0.588**
	(0.289)	(0.289)	(0.292)
Empl: other	0.071	0.081	0.070
	(0.332)	(0.337)	(0.340)
Income: 20–40k	–0.019	–0.013	–0.015
	(0.081)	(0.080)	(0.080)
Income: 40–60k	–0.078	–0.079	–0.064
	(0.158)	(0.159)	(0.163)
Income: 60–80k	–0.165	–0.171	–0.165
	(0.200)	(0.200)	(0.199)
Income: >80k	–0.339**	–0.334**	–0.313**
	(0.158)	(0.162)	(0.151)
Income: N/A	0.115	0.111	0.103
	(0.073)	(0.076)	(0.076)
Trust in mayor	0.836***	0.842***	0.838***
	(0.096)	(0.095)	(0.097)
<i>Municipal controls</i>			
Mun. damage: considerable	–0.207	–0.216	–0.227
	(0.248)	(0.257)	(0.243)
Mun. damage: minor	–0.220	–0.235	–0.234

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	(1) General IMC	(2) Territorial Mgmt	(3) Residual IMC
	(0.265)	(0.270)	(0.268)
Mun. damage: none	−0.141	−0.155	−0.168
	(0.332)	(0.329)	(0.339)
Home damage	0.091	0.098	0.096
	(0.082)	(0.081)	(0.083)
Population	−0.000	−0.000	−0.000*
	(0.000)	(0.000)	(0.000)
<i>Threshold parameters</i>			
κ_1	0.238	0.235	0.177
	(0.463)	(0.496)	(0.473)
κ_2	2.062***	2.062***	1.998***
	(0.460)	(0.494)	(0.466)
Observations	1,189	1,189	1,189

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Province-clustered SE in parentheses.

Reference categories: primary education, unemployed, income <20k, heavy municipal damage.